ANALOG ELETRONICS

LECTURE 1:

BIPOLAR JUNCTION TRANSISTOR (BJT)

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OUTLINE

- Introduction
- BJT construction
- Naming of BJT terminals
- Basic BJT Operation
- BJT Operating Modes
- Transistor Connection
- The BJT as an Amplifier

INTRODUCTION

- The transistor was invented in 1948 by J. Bardeen and W.H. Brattain of Bell Telephone Laboratories, U.S.A.;
- The **word Transistor** comes from the **words** transfer and resistor,
- Because an electrical current in a **transistor** is transferred across a resistor, a two-terminal component of electronics.
- The device logically belongs in the varistor family, and has the transconductance or transfer impedance of a device having gain.

INTRODUCTION

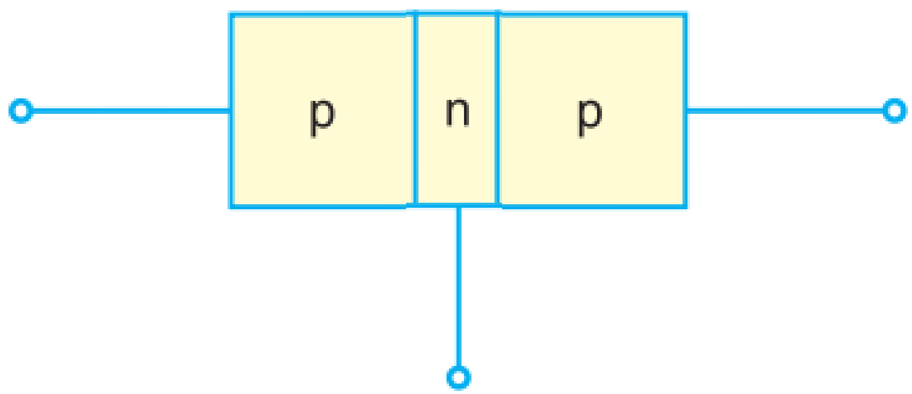
- Two basic types of transistors are the bipolar junction transistor (BJT), which we will begin to study in this lecture, and the field-effect transistor (FET) which we will cover in lecture 4.
- The BJT is used in two broad areas;
  - As a linear amplifier to boost or amplify an electrical signal
  - As an electronic switch.

Construction

- The Bipolar Junction Transistor is a three-layer semiconductor device consisting of either two n- and one p-type layers of material or two p- and one n-type layers of material.
- The term bipolar refers to the use of both holes and electrons as current carriers in the transistor structure.
- Based on its construction, a bipolar junction transistor can be classified into two types:
  - p-n-p transistor
  - n-p-n transistor

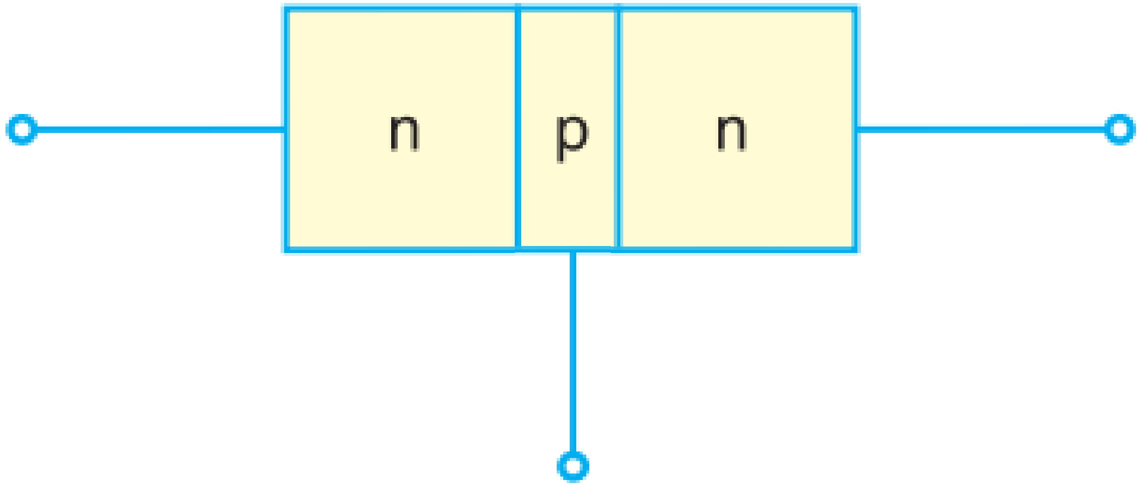
CONSTRUCTION

- A junction transistor consists of a silicon (or germanium) crystal in which a layer of n-type silicon is sandwiched between two layers of p-type silicon, can be referred as a **p-n-p transistor.**



Construction

- Alternatively, a junction transistor consists of a silicon (or germanium) crystal in which a layer of p-type silicon is sandwiched between two layers of n-type silicon, can be referred as a **n-p-n transistor.**



Construction

- In each type of transistor, the following points may be noted :

(i) These are two pn junctions. One junction is forward biased and the other is reverse biased.

- The forward biased junction has a low resistance path whereas
- A reverse biased junction has a high resistance path.
- The weak signal is introduced in the low resistance circuit and output is taken from the high resistance circuit.
- Therefore, a transistor may be regarded as a combination of two diodes connected back to back.

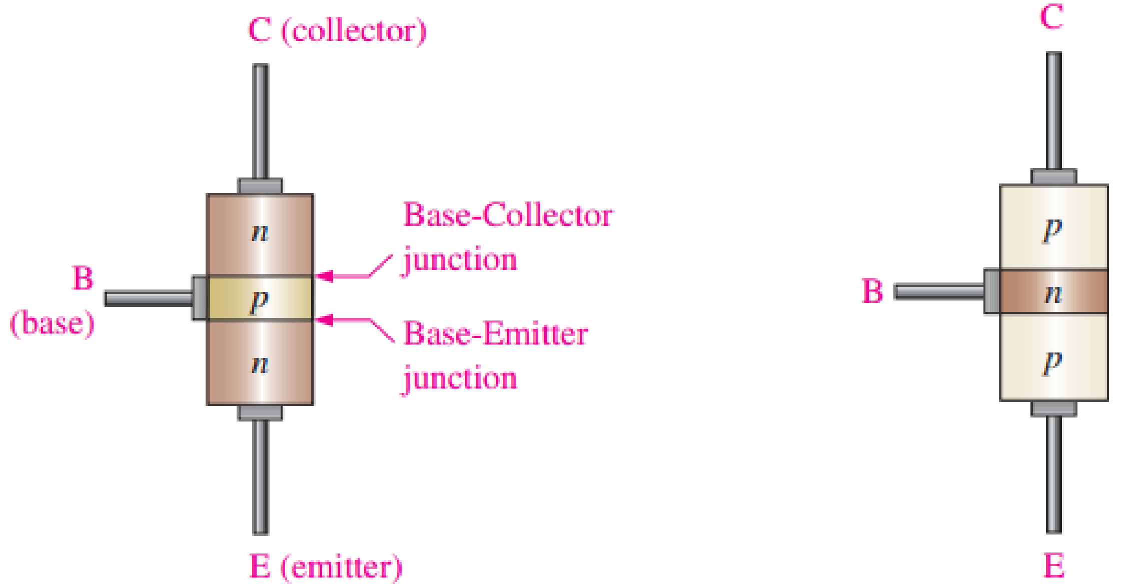
CONSTRUCTION

(ii) There are three terminals, one taken from each type of semiconductor.

(iii) The middle section is a very thin layer. This is the most important factor in the function of a transistor.

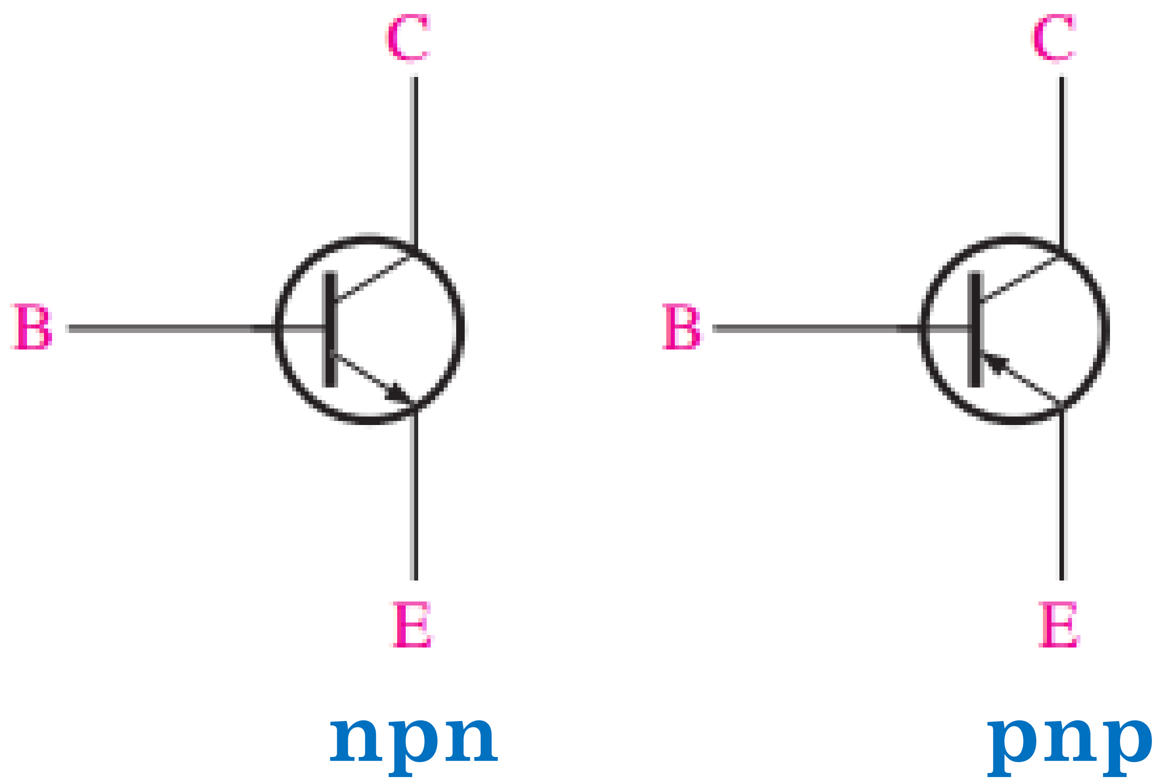
NAMING OF TRANSISTOR TERMINALS

- The three layer of transistor are Emitter (E), Base (B) and Collector (C).



- The base region is lightly doped and very thin compared to the heavily doped emitter and the moderately doped collector regions.

NAMING OF TRANSISTOR TERMINALS



Schematic symbols for the npn and pnp bipolar junction transistors

NAMING OF TRANSISTOR TERMINALS

1) Emitter:

- The section of one side that supplies carriers is called emitter.
- Emitter is always **forward biased** with respect to base so it can supply carriers.
- For "npn transistor" emitter supply holes to its junction.
- For "pnp transistor" emitter supply electrons to its junction.

NAMING OF TRANSISTOR TERMINALS

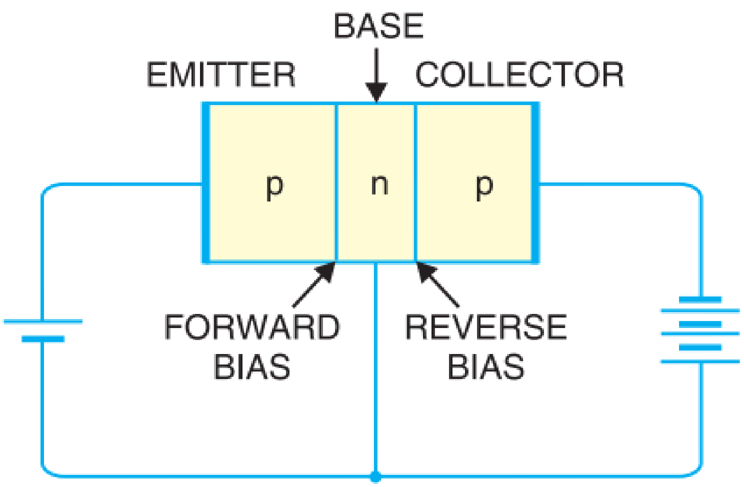
2) Collector:

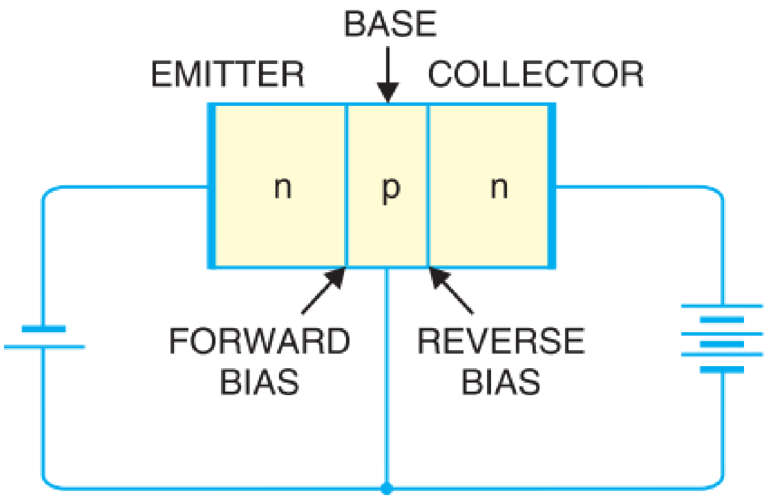
- The section on the other side that collects carrier is called collector.
- The collector is always **reversed biased** with respect to base.
- For "npn transistor" collector receives holes to its junction.
- For "pnp transistor" collector receives electrons to its junction.

NAMING OF TRANSISTOR TERMINALS

3) Base:

- The middle section which forms two pn junction between emitter and collector is called Base.





Some important factors to be remembered

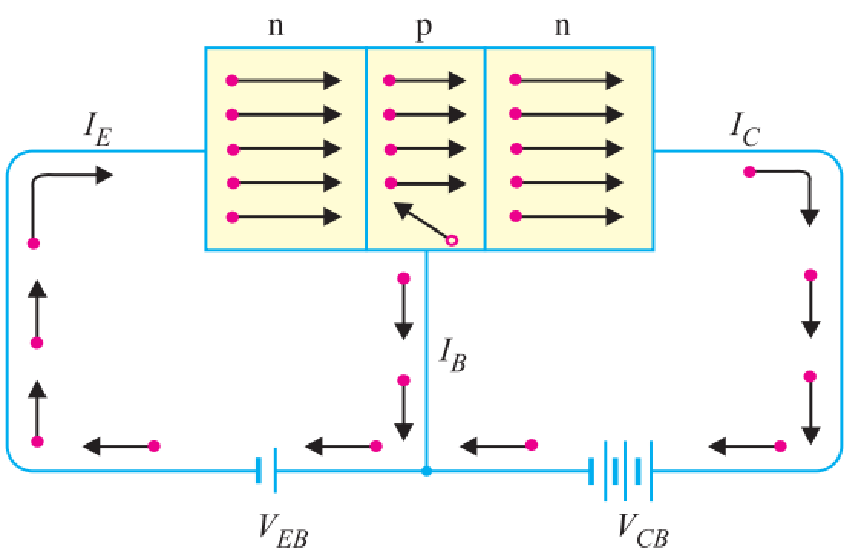
- The transistor has three region named emitter, base and collector.
- The Base is much thinner than other region.
- Emitter is heavily doped so it can inject large amount of carriers into the base.
- Base is lightly doped so it can pass most of the carrier to the collector.
- Collector is moderately doped.

Some important factors to be remembered

- The junction between emitter and base is called emitter-base junction(emitter diode) and junction between base and collector is called collector-base junction(collector diode).
- The emitter diode is always forward biased and collector diode is reverse biased.
- The resistance of emitter diode is very small(forward) and resistance of collector diode is high(reverse).

Basic BJT Operation

1) Working of npn transistor:



- Forward bias is applied to emitter base junction and reverse bias is applied to collector base junction.

- The forward bias causes the electrons in the n-type emitter to flow towards the base.
- This constitute emitter current, I_E

Basic BJT Operation

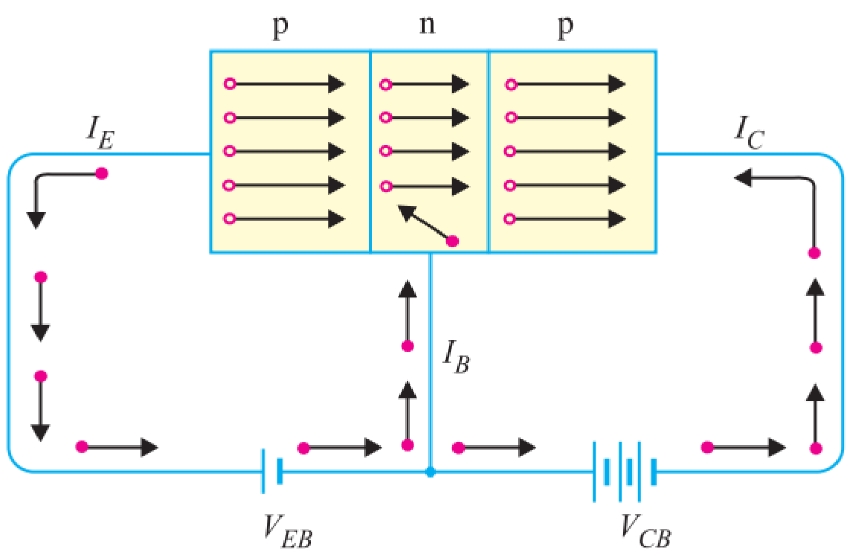
1) Working of npn transistor:

- As these electrons flow through the p-type base, they tend to combine with holes.
- As the base is lightly doped and very thin, therefore, only a few electrons (less than 5%) combine with holes to constitute base current I_B
- The remainder (more than 95%) cross over into the collector region to constitute collector current I_C.
- Thus;

$$I_E = I_B + I_C$$

Basic BJT Operation

2) Working of pnp transistor:



- Forward bias is applied to emitter base junction and reverse bias is applied to collector base junction.

- The forward bias causes the holes in the p-type emitter to flow towards the base.
- This constitute emitter current, I_E

Basic BJT Operation

2) Working of pnp transistor:

- As these holes cross into n-type base, they tend to combine with the electrons.
- As the base is lightly doped and very thin, therefore, only a few holes (less than 5%) combine with the electrons.
- The remainder (more than 95%) cross into the collector region to constitute collector current I_C.
- It may be noted that current conduction within pnp transistor is by holes.

BJT Operating Modes

- Active Mode;
  - Base-Emitter junction is forward
  - Base-Collector junction is reverse biased.

- Saturation Mode;
  - Base-Emitter junction is forward
  - Base-Collector junction is forward biased.

- Cut-off Mode;
  - Both junctions are reverse biased.

Transistor Connection

- Transistor can be connected in a circuit in following three ways;

1) Common Base

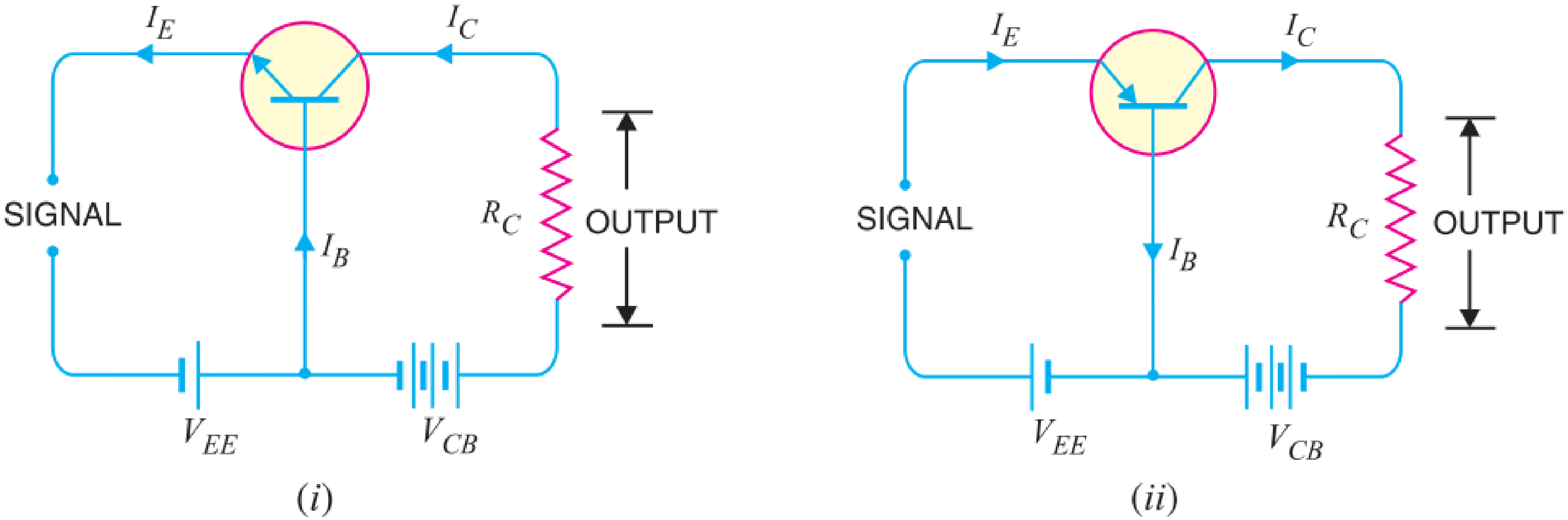
2) Common Emitter

3) Common Collector

Transistor Connection

1)**Common Base Connection**

- The base is common to both the input and output sides of the configuration.



Transistor Connection

1) **Common Base Connection**

- The Current amplification factor (α) :

- The ratio of change in collector current to the change in emitter current at constant V_{CB} is known as current amplification factor, α.

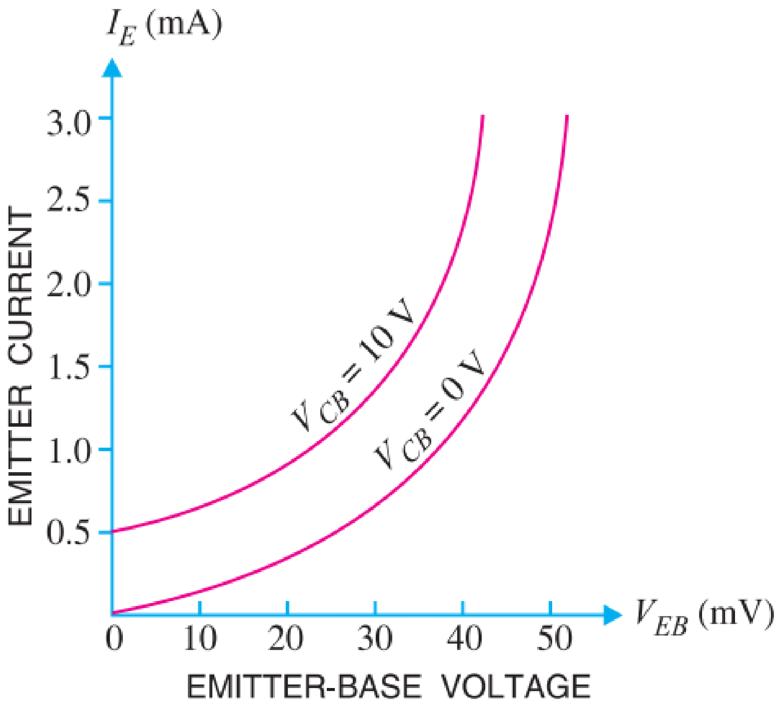
$$\alpha = \frac{\Delta I_C}{\Delta I_E} \text{ at Constant } V_{CB}$$

- Practical value of α is less than unity, but in the range of 0.9 to 0.99

Transistor Connection

1) Common Base Connection

- Characteristics of common base configuration;
- **Input Characteristics:**

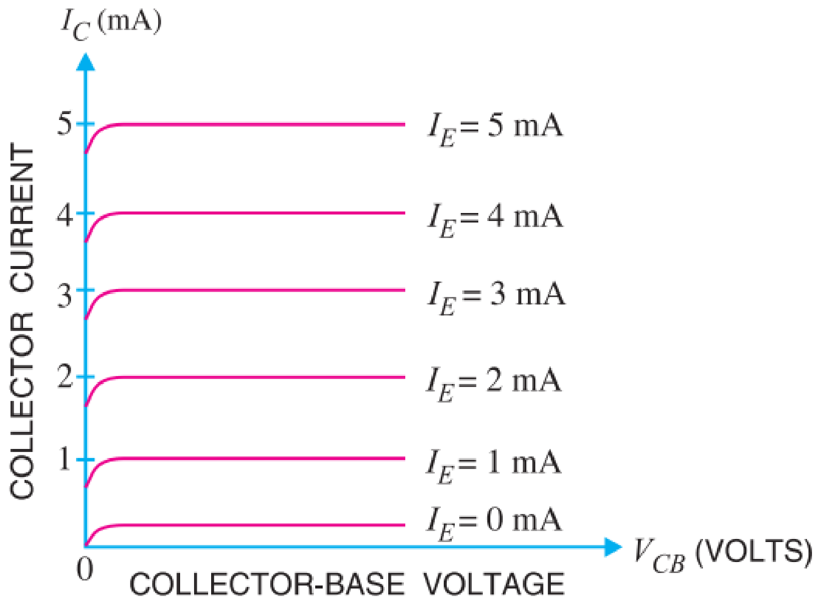


- I_E vs V_{BE} curve at constant V_{CB} is called input characteristics.
- I_E increases rapidly with V_{BE}
- It means input resistance is very small.
- I_E is almost independent of V_{CB}.

Transistor Connection

1)Common Base Connection

- Characteristics of common base configuration;
- **Output Characteristics**:



- I_C vs V_{CB} curve at constant I_E
- The I_C varies with V_{CB} only at very low voltages (< 1V).
- The transistor is never operated in this region.
- I_C varies linearly with V_{CB} ,only when V_{CB} is very small.
- As, V_{CB} increases, I_C becomes constant.

Transistor Connection

1) Common Base Connection

- Characteristics of common base configuration;

- **Input resistance**:

The ratio of change in emitter-base voltage (ΔV_{BE}) to the resulting change in emitter current (ΔI_E) at constant collector-base voltage (V_{CB})

$$\text{Input resistance}, r_i = \frac{\Delta V_{BE}}{\Delta I_E} \text{ at contant } V_{CB}$$

Transistor Connection

1) Common Base Connection

- Characteristics of common base configuration;

- **Output resistance**:

The ratio of change in collector-base voltage (ΔV_{CB}) to the resulting change in collector current (ΔI_C) at constant emitter current (I_E) .

$$\text{Input resistance}, r_o = \frac{\Delta V_{CB}}{\Delta I_C} \text{ at constant } I_E$$

Transistor Connection

Example 1: *In a common base connection,* $I_E =$ *1mA,* $I_C =$ *0.95mA. Calculate the value of* I_B.

Solution:

Using the relation, $I_E = I_B + I_C$ or $1 = I_B + 0.95$

$\therefore\ I_B = 1 - 0.95 =$ 0.05 mA

Example 2: *In a common base connection, current amplification factor is 0.9. If the emitter current is 1mA, determine the value of base current.*

Solution:

Here, $\alpha = 0.9$, $I_E = 1\text{mA}$

Now $\alpha = \frac{I_C}{I_E}$

or $I_C = \alpha\, I_E = 0.9 \times 1 =$ 0.9 mA

Also $I_E = I_B + I_C$

$\therefore$ Base current, $I_B = I_E - I_C = 1 - 0.9 =$ 0.1 mA

TRANSISTOR CONNECTION

Example 3: *In a common base connection,* I_C*= 0.95mA and* I_B *= 0.05 mA. Find the value of* α

Solution:

We know $I_E = I_B + I_C = 0.05 + 0.95 = 1$ mA

$\therefore$ Current amplification factor, $\alpha = \frac{I_C}{I_E} = 0.95$

Example 4: *In a common base connection,* α *= 0.95. The voltage drop across 2 kΩ resistance which is connected in the collector is 2V. Find the base current.*

Solution:

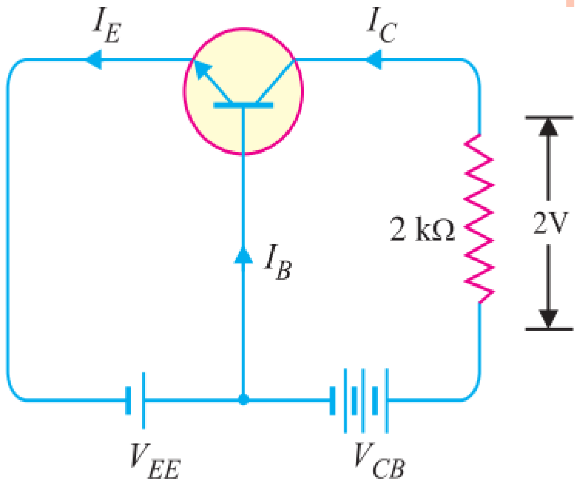
The voltage drop across RC(=2 kΩ) is 2V.

$I_C = \frac{2\ V}{2\ k\Omega} = 1\text{mA}$,

Now, $\alpha = \frac{I_C}{I_E}$ or $I_E = \frac{I_C}{\alpha} = \frac{1\text{mA}}{0.95} = 1.05\ mA$

Using the relation, $I_E = I_B + I_C$ or $1.05 = I_B + 1$

$\therefore\ I_B = 1.05 - 1 = 0.05$ mA



Transistor Connection

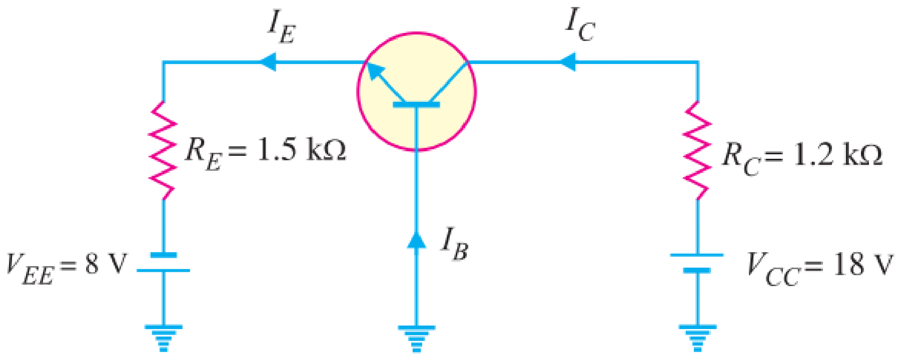
Example 5: *In For the common base circuit shown in Fig below, determine I_C and V_{CB}. Assume the transistor to be of silicon.*

Solution: Since the transistor is of silicon, V_{BE} = 0.7V. Applying Kirchhoff's voltage law to the emitter-side loop, we get;

$$V_{EE} = I_E R_E + V_{BE}$$

$$I_E = \frac{V_{EE} - V_{BE}}{R_E} = \frac{8V - 0.7V}{1.5\text{k}} = 4.87\text{mA}$$

$$\therefore \; I_C \approx I_E = 4.87\text{mA}$$



Applying Kirchhoff's voltage law to the collector-side loop, we have,

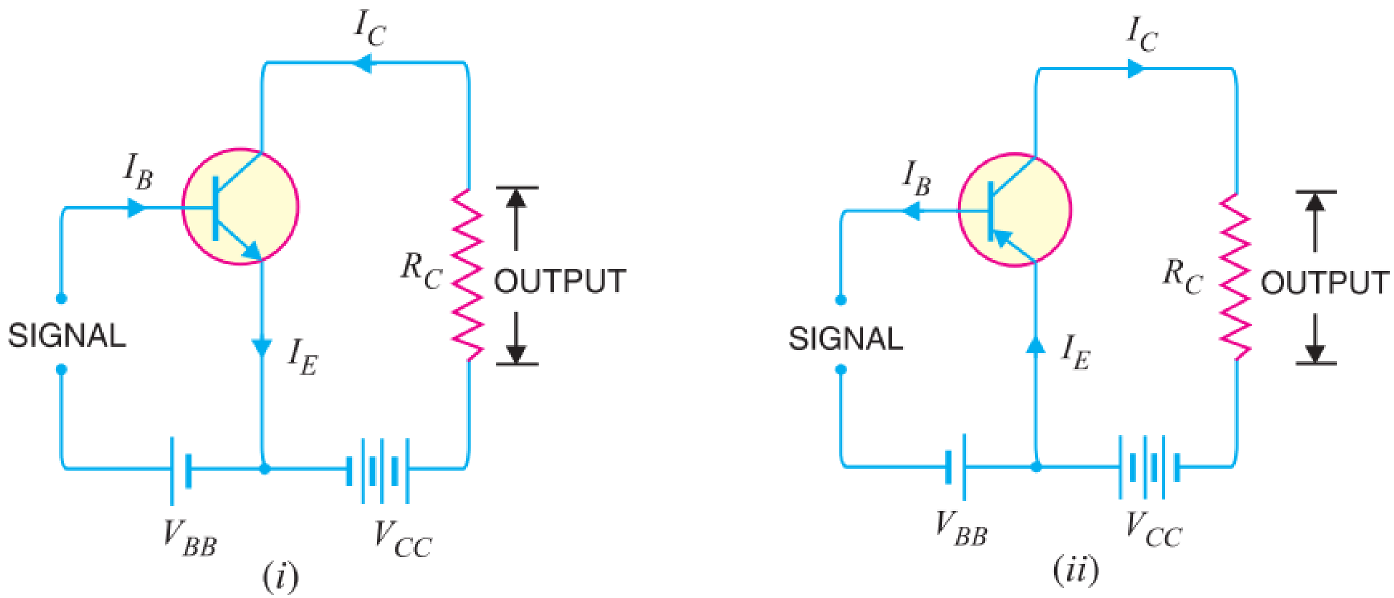
$$V_{CC} = I_C R_C + V_{CB}$$

$$V_{CB} = V_{CC} - I_C R_C = 18V - 4.87m \; x \; 1.2k = 12.16V$$

TRANSISTOR CONNECTION

2)**Common Emitter Connection**

- The emitter is common to both the input and output sides of the configuration.
- Input is applied between base and emitter and output is taken from the collector and emitter.



Transistor Connection

2) **Common Emitter Connection**

- Base current amplification factor (β):

- The ratio of change in collector current (ΔI_C) to the change in base current (ΔI_B) is known as base current amplification factor.

$$\beta = \frac{\Delta I_C}{\Delta I_B}$$

- The value of β ranges from 20 to 500.
- This type of connection is frequently used as it gives appreciable current gain as well as voltage gain.

Transistor Connection

2) Common Emitter Connection

- Relation between β and α :

$$\beta = \frac{\Delta I_C}{\Delta I_B} \text{ and } \alpha = \frac{\Delta I_C}{\Delta I_E}$$

Now $\quad I_E = I_C + I_B$

or $\quad \Delta I_E = \Delta I_C + \Delta I_B$

or $\quad \Delta I_B = \Delta I_E - \Delta I_C$

$$\beta = \frac{\Delta I_C}{\Delta I_E - \Delta I_C} = \frac{\Delta I_C / \Delta I_E}{\Delta I_E / \Delta I_E - \Delta I_C / \Delta I_E} = \frac{\alpha}{1 - \alpha}$$

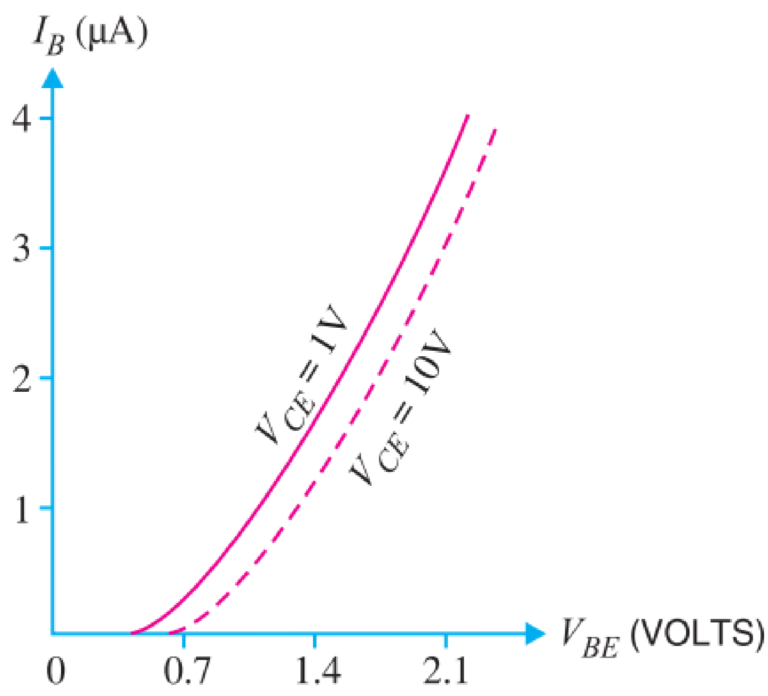
$$\beta = \frac{\alpha}{1 - \alpha}$$

Transistor Connection

2) Common Emitter Connection

- Characteristics of common emitter configuration :

- **Input characteristics**:

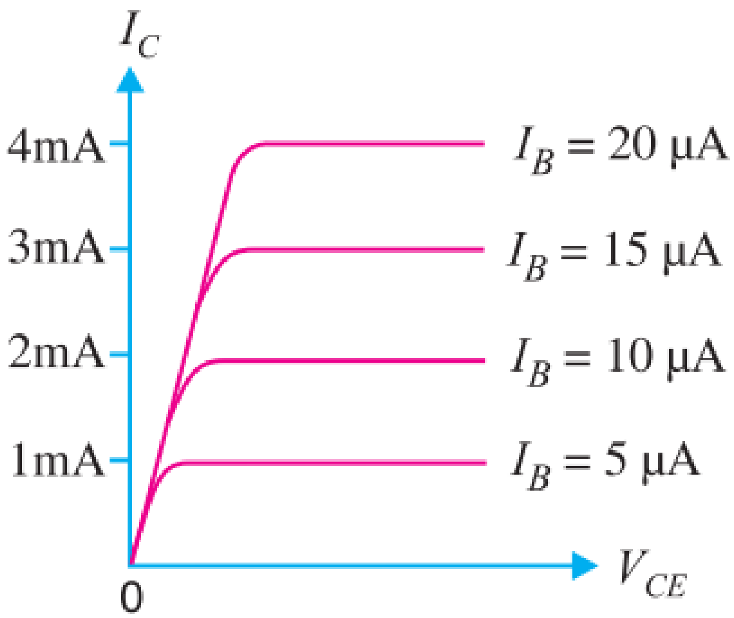


- V_{BE} vs I_B curve is called input characteristics.
- I_B increases rapidly with V_{BE}.
- It means input resistance is very small.
- I_E almost independent of V_{CE}.
- I_B is of the range of micro amps.

Transistor Connection

2) Common Emitter Connection

- Characteristics of common emitter configuration :
- **Output characteristics :**



- V_{CE} vs I_C curve is called output characteristics.
- I_C varies linearly with V_{CE} ,only when V_{CE} is very small.
- As, V_{CE} increases, I_C becomes constant.

2)Common Emitter Connection

- Characteristics of common emitter configuration :
- **Input resistance**:
- The ratio of change in base-emitter voltage (ΔV_{BE}) to the change in base current (ΔI_B) at constant V_{CE}.

$$\text{Input resistance}, r_i = \frac{\Delta V_{BE}}{\Delta I_B} \text{ at constant } V_{CE}$$

Transistor Connection

2)**Common Emitter Connection**

- Characteristics of common emitter configuration :
- **Output resistance**:
- The ratio of change in collector-emitter voltage (ΔV_{CE}) to the change in collector current (ΔI_C) at constant I_B

$$\text{Output resistance}, r_o = \frac{\Delta V_{CE}}{\Delta I_C} \text{ at constant} I_B$$

Transistor Connection

Example 6:

Find the value of β if (i) α = 0.9 (ii) α = 0.98 (iii) α = 0.99.

Solution:

(i) $\beta = \frac{\alpha}{1-\alpha} = \frac{0.9}{1-0.9} = 9$

(ii) $\beta = \frac{\alpha}{1-\alpha} = \frac{0.98}{1-0.98} = 49$

(iii) $\beta = \frac{\alpha}{1-\alpha} = \frac{0.99}{1-0.99} = 99$

Example 7:

Calculate I_E in a transistor for which β= 50 and I_B = 20 μA

Solution:

Here β = 50, I_B = 20μA = 0.02 mA

$$\beta = \frac{I_C}{I_B}$$

$$I_C = \beta I_B = 50\ x\ 0.02\ mA = 1\text{mA}$$

Using the relation, $I_E = I_C + I_B = 0.02 + 1 = 1.02$ mA

Transistor Connection

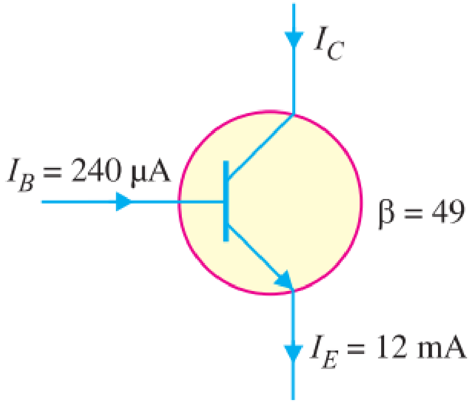
Example 8:

Find the α rating of the transistor shown in Fig. Hence determine the value of I_C using both α and β rating of the transistor.

Solution:

$$\alpha = \frac{\beta}{1+\beta} = \frac{49}{1+49} = 0.98$$

The value of I_C can be found by using either α or β rating as under :



$I_C = \alpha\, I_E = 0.98\ x\ 12\ mA = 11.76\ mA$

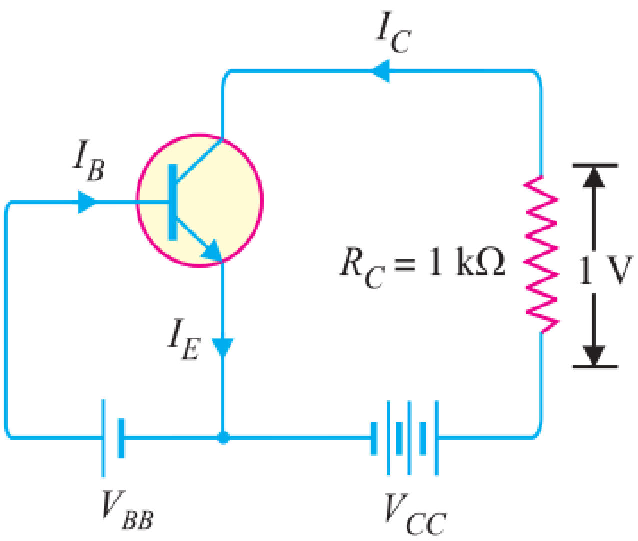
Also $I_C = \beta\, I_B = 49\ x\ 240\ \mu A = 11.76\ mA$

Transistor Connection

Example 9:

For a transistor, β= 45 and voltage drop across 1kΩ which is connected in the collector circuit is 1 volt. Find the base current for common emitter connection.

Solution:



$$I_C = \frac{1V}{1k\Omega} = 1\,mA$$

Now $\beta = \frac{I_C}{I_B}$

∴ $I_B = \frac{I_C}{\beta} = \frac{1}{45} = 0.022\,mA$

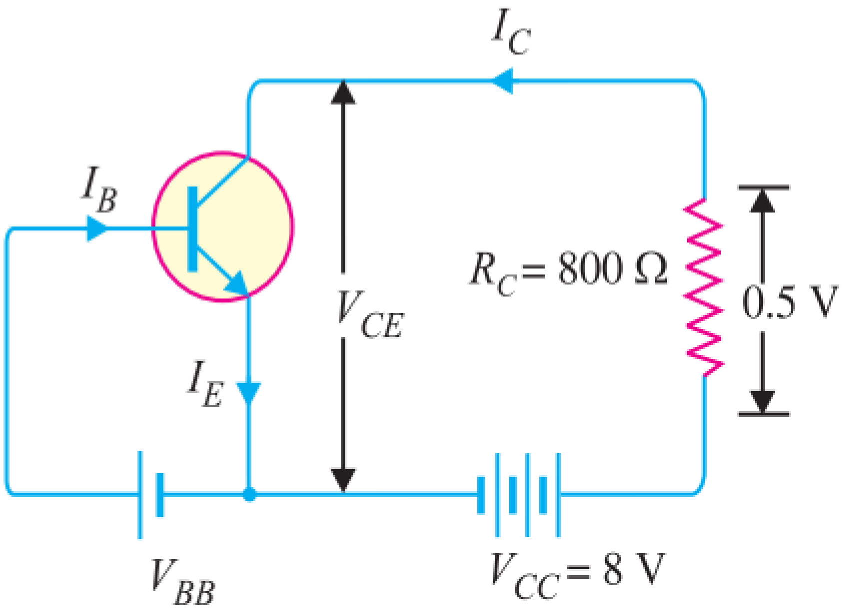
Transistor Connection

Review Questions

1. A transistor is connected in common emitter (CE) configuration in which collector supply is 8V and the voltage drop across resistance R_C connected in the collector circuit is 0.5V. The value of R_C = 800 Ω. If α = *0.96, determine :*

(i) collector-emitter voltage

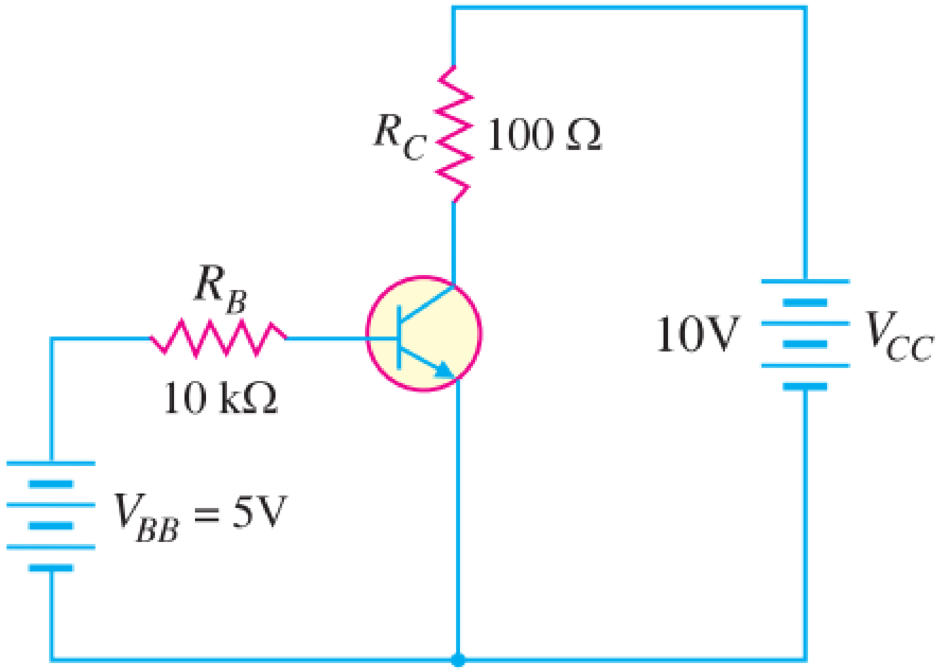
(ii) base current



Transistor Connection

Review Questions

2. Determine V_{CB} in the transistor circuit shown in Fig. The transistor is of silicon and has $\beta = 150$.



Transistor Connection

Review Questions

3. *In a transistor,* $I_B = 68\ \mu A$, $I_E = 30\ mA$ *and* $\beta = 440$. *Determine the α rating of the transistor. Then determine the value of* I_C *using both the α rating and β rating of the transistor.*

4. *A transistor has the following ratings :* $I_C\ (max) = 500\ mA$ *and* $\beta_{max} = 300$. *Determine the maximum allowable value of* I_B *for the device.*

5. *What is the significance of arrow in the transistor symbol ?*

6. *Why is collector wider than emitter and base ?*

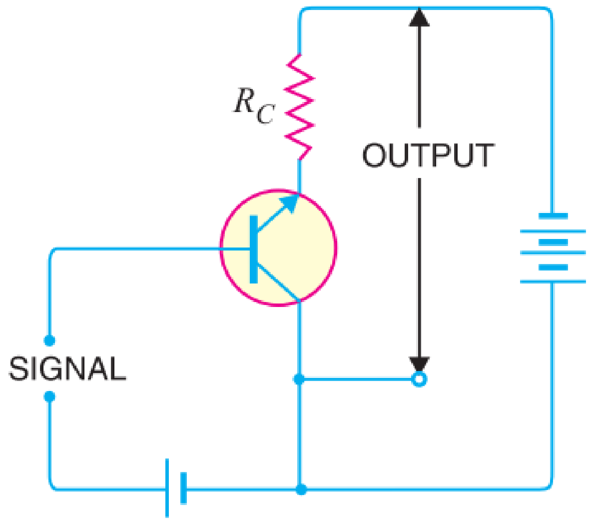
7. *Why is collector current slightly less than emitter current ?*

8. *Why is base made thin ?*

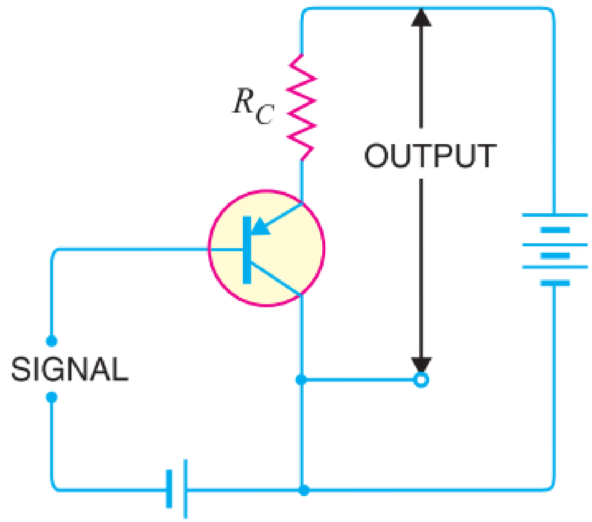
TRANSISTOR CONNECTION

3)**Common Collector Connection**

- The collector is common to both the input and output sides of the configuration.
- Input is applied between base and collector while output is taken between the emitter and collector.



(i)



(ii)

Transistor Connection

3) **Common Collector Connection**

- Current amplification factor (γ)

- The ratio of change in emitter current (ΔI_E) to the change in base current (ΔI_B) is known as current amplification factor in common collector (CC) arrangement.

$$\gamma = \frac{\Delta I_E}{\Delta I_B}$$

- Input current is the base current I_B and output current is the emitter current I_E.
- Current gain is same as in the common emitter circuit as $\Delta I_E \approx \Delta I_C$
- However, its voltage gain is always less than 1.

Transistor Connection

3)Common Collector Connection

- Relation between γ and α

$$\gamma = \frac{\Delta I_E}{\Delta I_B} \text{ and } \alpha = \frac{\Delta I_C}{\Delta I_E}$$

Now

$$\Delta I_B = \Delta I_E - \Delta I_C$$

$$\gamma = \frac{\Delta I_E}{\Delta I_E - \Delta I_C}$$

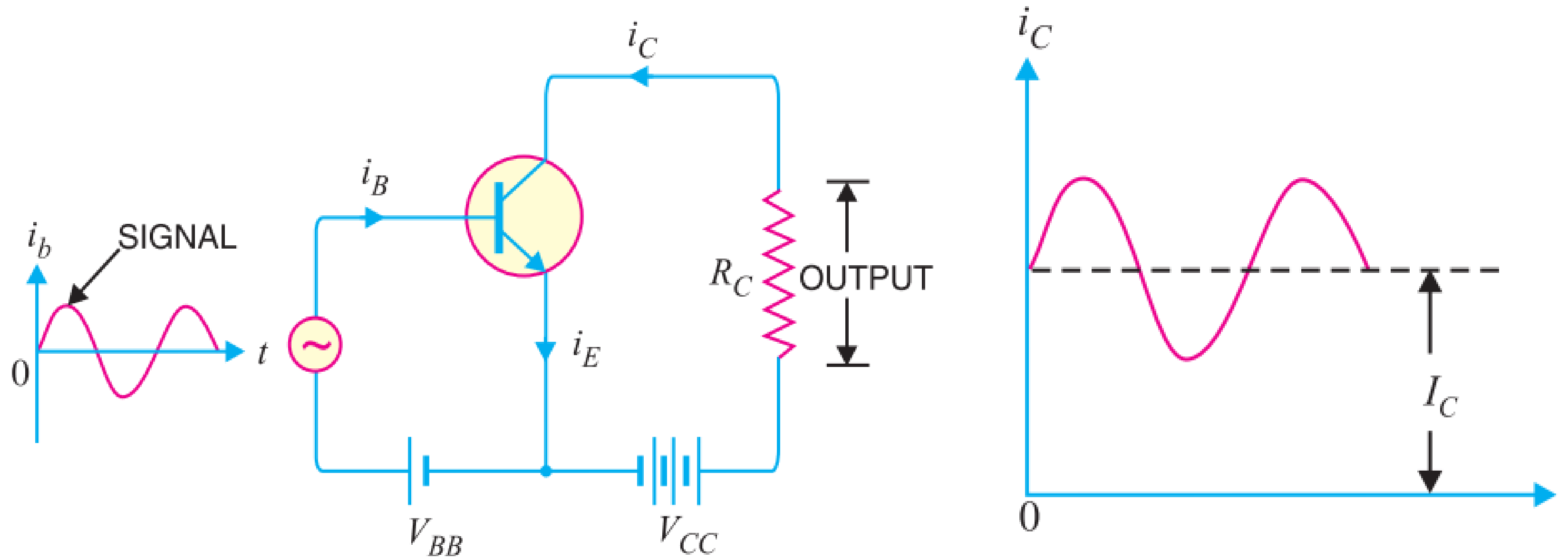
$$\gamma = \frac{\frac{\Delta I_E}{\Delta I_E}}{\frac{\Delta I_E}{\Delta I_E} - \frac{\Delta I_C}{\Delta I_E}} = \frac{1}{1 - \alpha}$$

$$\gamma = \frac{1}{1 - \alpha}$$

Commonly Used Transistor Connection

- Out of the three transistor connections, the **common emitter circuit** is the most efficient.
- It is used in about 90% to 95% of all transistor applications.
- The main reasons for the widespread use of this circuit arrangement are :
  - High current gain- range from 20 to 500
  - High voltage and power gain-due to high current gain
  - Moderate output to input impedance ratio- about 50

Transistor as an Amplifier in CE Arrangement



- This is the common emitter npn amplifier circuit.
- A battery V_{BB} is connected in the input circuit in addition to the signal voltage.
- This d.c voltage is known as *bias voltage* and its magnitude is such that it always keeps the emitter-base junction forward biased regardless of the polarity of the signal source.

Transistor as an Amplifier in CE Arrangement

Operation

- During the positive half-cycle of the signal, the forward bias across the emitter-base junction is increased.
- Therefore, more electrons flow from the emitter to the collector via the base.
- This causes an increase in collector current.
- The increased collector current produces a greater voltage drop across the collector load resistance R_C
- However, during the negative half-cycle of the signal, the forward bias across emitter-base junction is decreased.
- Therefore, collector current decreases. This results in the decreased output voltage (in the opposite direction).
- Hence, an amplified output is obtained across the load.

NEXT SESSION …BJT BIAS CIRCUITS